Chapter 6

APPLICATIONS OF DIFFERENTIATION

Learning Objectives

After studying this chapter, the students will be able to understand

- the concepts of demand and supply
- meaning and uses of cost, revenue and profit function
- average and marginal concepts
- elasticity of demand and supply
- relationship among average revenue, marginal revenue and elasticity of demand
- application of increasing and decreasing functions
- application of maxima and minima
- concept of Economic Order Quantity
- concepts of partial differentiation
- applications of partial derivatives in Economics
- partial elasticities of demand

Introduction


Leonhard Euler

Modern economic theory is based on both differential and integral calculus. In economics, differential calculus is used to compute marginal cost, marginal revenue, maxima and minima, elasticities, partial elasticities and also enabling economists to predict maximum profit (or) minimum loss in a specific condition. In this chapter, we will study about some important concepts and applications of differentiation in business and economics.

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Euler's theorem in Economics

If factors of production function are paid as factors times of their marginal productivities, then the total factor payment is equal to the degree of homogeneity times the production function.

6.1 Applications of Differentiation in Business and Economics

In an economic situation, consider the variables are price and quantity. Let p be the unit price in rupees and x be the production (output / quantity) of a commodity demanded by the consumer (or) supplied by the producer.

6.1.1 Demand, supply, cost, revenue and profit functions

Demand function

In a market, the quantity of a commodity demanded by the consumer depends on its price. If the price of the commodity increases, then the demand decreases and if the price of the commodity decreases, then the demand increases.

The relationship between the quantity and the unit price of a commodity demanded by consumer is called as **demand function**

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